World Models

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VPFlow: Value-Predictive Planning

Hu and Xu [1] introduce VPFlow, a framework for learning latent world models optimized for goal-conditioned planning rather than accurate state prediction. Their central argument is that planning does not require imagined states to match the true future states, provided they preserve the values needed to evaluate candidate actions. They propose jointly training the encoder, latent dynamics model, and goal-conditioned value function using hindsight Monte Carlo returns as the sole supervision for the learned representations and dynamics.

Hindsight Monte Carlo Return Supervision

In offline goal-conditioned reinforcement learning, recorded trajectories can be retrospectively relabeled by selecting any future state as a goal. Given a trajectory $(s_0, a_0, s_1, \ldots, s_T)$, let $g = s_{t+d}$ denote a goal reached after d steps. Under a sparse reward of -1 until reaching the goal and 0 upon reaching it, the realized discounted return is

$$R(s_t, g) = -\sum_{k=0}^{d-1} \gamma^k = \frac{1-\gamma^d}{\gamma - 1},$$

where $\gamma \in [0, 1)$ is the discount factor. This provides a closed-form training target based on the temporal distance to the goal, eliminating the need for temporal-difference bootstrapping or target networks. Importantly, the target reflects the return along the observed trajectory, not necessarily the optimal goal-reaching return.

Value-Predictive Latent Dynamics

VPFlow jointly learns an encoder E_ϕ, an action-conditioned latent dynamics model f_ϕ, and a goal-conditioned value function V_ϕ, with ϕ denoting their collective parameters. Given an initial state s_t, goal g, and recorded action sequence $a_{t:t+H-1}$ of horizon H, define the initial and goal representations as $z_t = E_\phi(s_t)$ and $z_g = E_\phi(g)$. The dynamics model predicts

$$\hat{z}_{t+h} = f_\phi(z_t, a_{t:t+h-1}), \qquad h = 1, \ldots, H,$$

with $\hat{z}_t = z_t$. Rather than matching these predictions to the encoded future states, the model minimizes

$$\mathcal{L}_{\mathrm{VP}}(\phi) = \mathbb{E}\left[\frac{1}{H+1}\sum_{h=0}^{H} M_{t+h}\left(V_\phi(\hat{z}_{t+h}, z_g) - R(s_{t+h}, g)\right)^2\right].$$

Here h indexes the prediction horizon, $V_\phi(\hat{z}_{t+h}, z_g)$ is the predicted goal-reaching value of the imagined latent state, and $R(s_{t+h}, g)$ is the closed-form Monte Carlo return from the corresponding observed state s_{t+h} to goal g, calculated using the temporal distance from $t+h$ to the goal. The binary mask M_{t+h} equals one when the goal occurs at or after $t+h$ on the same recorded trajectory, and zero otherwise. The expectation is over trajectory segments sampled from the offline dataset and sampled goals. The factor $1/(H+1)$ averages the loss across prediction horizons.

Gradients update the encoder, dynamics model, and value function jointly. The key idea is **value equivalence**: imagined latent states need not coincide with their observed counterparts, provided they lie on the appropriate level sets of the goal-conditioned value function. This relaxes the state-prediction

requirement and allows the representation to emphasize information relevant to reaching goals. Moreover, variation in goal-conditioned return targets implicitly discourages representation collapse without requiring explicit regularizers such as VICReg or SIGReg.

Experiments demonstrate that adding latent-state prediction can dramatically reduce prediction error while degrading planning performance, even when explicit anti-collapse regularization is included. Thus, accurate latent-state prediction is not a prerequisite for effective planning, although matching values alone does not guarantee correct predictions for arbitrary unseen action sequences.

Events, Not Pixels: Where World Models Get the Return Wrong

Liu and Tang [2] investigate why visually accurate world models can nevertheless produce inaccurate return predictions. They introduce a transition-level diagnostic showing that return errors are concentrated around reward and termination events, even when these transitions exhibit relatively small pixel-level errors. Their findings suggest that accurate representation of reward-relevant events is more important for planning than uniformly improving visual reconstruction quality.

Transition-Level Error Attribution

The authors compare imagined trajectories with real trajectories generated from the same initial context and action sequence. A frozen detector D estimates rewards and termination from consecutive observations:

$$(\hat{r}_t, \hat{d}_t) = D(\hat{o}_t, \hat{o}_{t+1}, a_t),$$

where $\hat{o}_t$ is the predicted observation, a_t is the action, and $\hat{r}_t$ and $\hat{d}_t$ are the estimated reward and termination indicator. These predictions determine the imagined cumulative return $\hat{G}$, which is compared with the true return G.

To identify where return errors originate, the authors selectively replace predicted transition pairs $(\hat{o}_t, \hat{o}_{t+1})$ with their ground-truth counterparts (o_t, o_{t+1}) before evaluating the detector. Importantly, subsequent predictions are not regenerated, so the intervention isolates the contribution of the selected transitions to the estimated return. For a substitution mask m, the relative reduction in return error is

$$R(m) = 1 - \frac{\mathbb{E}[|\hat{G}^m - G|]}{\mathbb{E}[|\hat{G} - G|]},$$

where $\hat{G}^m$ denotes the return after substitution.

The authors compare substitutions around true reward and termination events with an equal number of randomly selected transitions, as well as complementary non-event substitutions. Event-targeted corrections remove substantially more return error than random corrections or corrections targeting the largest pixel-level errors (MSE, SSIM, or LPIPS). Thus, the transitions at which return errors manifest are not reliably identified by conventional visual prediction metrics.

Missing Versus Mistimed Events

The authors investigate whether world models fail to represent reward events or merely predict them at incorrect times. They measure event recall while allowing predictions to match true events within a temporal tolerance τ. If events are simply mistimed, recall should improve substantially as τ increases. Instead, recall remains low for several deterministic world models, suggesting that many reward-relevant events are absent from the generated trajectories rather than temporally displaced.

One possible explanation is that deterministic models trained with squared-error objectives approximate conditional means, potentially averaging over distinct future outcomes and suppressing discrete events. To investigate this hypothesis, the authors compare a deterministic MSE world model with a diffusion-based model trained on the same data. The diffusion model recovers many missing events and improves return prediction. However, training the same diffusion network architecture with a deterministic squared-error objective produces comparable improvements. This suggests that the gains arise primarily from the network architecture rather than stochastic sampling or the diffusion objective itself.

Consequences for Planning

Finally, the authors examine whether correcting event predictions improves the ranking of candidate action sequences. Starting from a common environment state, they compare model-predicted returns across candidate plans with their true returns. Replacing predicted event transitions with real observations improves plan-ranking accuracy and reduces regret relative to randomly selected substitutions. The key observation is that world models intended for planning should preserve reward-relevant events, even when these events occupy only a small fraction of the imagined trajectory.

Understanding the Limits of Latent World Models

Laiz and Schneider [3] investigate whether latent world models that perform well on control tasks actually learn the underlying dynamics of their environments. They formalize the notion of a **faithful latent world model** and introduce diagnostics that distinguish accurate dynamics learning from successful task performance. Their analysis of DreamerV3, TD-MPC2, and LeWM highlights the difficulties of recovering dynamical state information from pixels and learning transitions under nonlinear observation mappings.

Faithful Latent World Models

Consider a deterministic dynamical system with state s_t, action a_t, transition function f, and observation function g:

$$s_{t+1} = f(s_t, a_t), \qquad x_t = g(s_t).$$

A latent world model consists of an encoder h, producing representations $z_t = h(x_{\leq t}, a_{<t})$, and a learned transition function $\hat{f}$, predicting $\hat{z}_{t+1} = \hat{f}(z_t, a_t)$. Writing $r = h \circ g$ for the state-to-latent mapping, the authors define faithfulness through two conditions:

1. **Dynamics equivariance:** $\hat{f}(r(s), a) = r(f(s, a))$. Evolving the state and then encoding it must agree with evolving its latent representation directly.
2. **State sufficiency:** r is injective, ensuring that the representation preserves all information about the physical state.

These conditions do not require latent coordinates to correspond directly to physical variables. Any invertible representation is permissible, provided the learned dynamics are consistent with the underlying system. Injectivity additionally excludes collapsed or task-specific representations that discard physical-state information.

Evaluating Faithfulness

The authors introduce **latent rollout consistency** (LRC) to measure whether autoregressive predictions remain consistent with representations inferred from realized trajectories. Starting from an encoded state, the model predicts $\hat{z}_H$ by repeatedly applying $\hat{f}$ under a recorded sequence of actions. This is compared with z_H, obtained by encoding the corresponding observed trajectory:

$$\mathrm{LRC}(H) = R^2(z_H, \hat{z}_H).$$

The metric aggregates coordinate-wise R^2 values using variance weights. High LRC indicates that the learned dynamics predict the model's own representations accurately, but does not guarantee that those representations preserve the physical state. The authors therefore complement LRC with nonlinear probes trained to recover physical-state variables from frozen latents.

These diagnostics reveal that **successful control does not imply faithful dynamics**. Reward-driven models can achieve high returns despite inaccurate open-loop predictions, potentially because their policies and value functions compensate for deficiencies in the explicit dynamics model or exploit task-specific representations.

Learning Dynamics from Observations

The authors identify two challenges in learning faithful dynamics from pixels.

1. **Recovering velocities.** A single image generally reveals position but not velocity, even though both are required for a Markovian representation of a mechanical system. Velocity must instead be inferred from temporal observations. Providing ground-truth velocities alongside pixel inputs substantially improves latent predictions for some models, demonstrating that state estimation is an important bottleneck. However, improvements in predictive consistency do not necessarily translate into better control.

2. **Learning under nonlinear observation mappings.** To isolate the difficulty of representation learning from information loss, the authors replace physical-state observations with invertible nonlinear transformations $x = g(s)$. The resulting dynamics in observation space are

$$x_{t+1} = g\left(f(g^{-1}(x_t), a_t)\right).$$

   Although these transformations preserve all state information, increasingly complex mappings degrade both latent prediction and control. Thus, information sufficiency does not guarantee that a representation and its associated dynamics can be learned effectively.

The central implication is that accurate world modeling requires both recovering a sufficient representation of the environment and learning dynamics that are predictable in the resulting coordinates. The experiments do not fully disentangle these two sources of error, but demonstrate that optimizing for task reward alone need not produce either property.

world2vec

Schmidt *et al.* [4] introduce world2vec, a self-supervised framework for jointly learning latent representations and action-conditioned dynamics from observations, without reward or reconstruction supervision. The approach combines multi-step latent prediction, continuous-time dynamics, and representation regularization to learn world models that reproduce the evolution of the underlying system.

Continuous-Time Latent Dynamics Matching

Consider a physical system with state $s(t)$, action $a(t)$, and observation $x(t) = g(s(t))$, governed by $\dot{s}(t) = f(s(t), a(t))$. World2vec learns an encoder h and latent dynamics model $\hat{f}_\theta$ such that

$$z(t) = h(x(t)), \qquad \dot{z}(t) = \hat{f}_\theta(z(t), a(t)).$$

The objective is for the integrated latent dynamics to reproduce the encoding of future observations. To account for partial observability, the encoder takes a short observation history X_t rather than a single frame.

The ODE is approximated using one Euler step per observation interval Δt:

$$\hat{z}_{t,0} = h(X_t), \qquad \hat{z}_{t,j+1} = \hat{z}_{t,j} + \Delta t\, \hat{f}_\theta(\hat{z}_{t,j}, a_{t+j}).$$

The encoder and dynamics model are jointly trained using

$$\mathcal{L}_{\text{world2vec}} = \mathbb{E}\left[\sum_{j=1}^{K} w_j \|\hat{z}_{t,j} - h(X_{t+j})\|_2^2\right] + \lambda \mathcal{R},$$

where K is the rollout horizon, w_j are horizon weights (typically $1/K$), and $\mathcal{R}$ prevents representation collapse. Because each step uses the preceding predicted latent, the objective penalizes accumulated prediction errors.

For fixed Δt, Euler integration is equivalent to residual next-state prediction, $\hat{z}_{t+1} = z_t + F_\theta(z_t, a_t)$, with $F_\theta = \Delta t\, \hat{f}_\theta$. The experiments compare this formulation against direct next-state prediction, but not a residual model, so they do not isolate an advantage of continuous-time dynamics beyond residual parameterization. The ODE formulation also allows the integration step size to be changed at inference.

Representation Regularization and Identifiability

World2vec considers two approaches to preventing representation collapse. SIGReg, as in LeJEPA, encourages embeddings to follow an isotropic Gaussian distribution. Dynamics Contrastive Learning (DCL) instead encourages separation between embeddings of different observations.

For prediction horizon j, define the squared prediction error $d_{t,j}$ and negative-sample similarity sum S_t:

$$d_{t,j} = \|\hat{z}_{t,j} - h(X_{t+j})\|_2^2, \qquad S_t = \sum_{i=1}^{M} \exp\left(-\|h(X_t) - h(X_{m_i})\|_2^2\right),$$

where $X_{m_1}, \ldots, X_{m_M}$ are randomly sampled negative observation histories. Combining the prediction loss with DCL regularization gives

$$\mathcal{L}_{\text{DCL}} = \mathbb{E}\left[\sum_{j=1}^{K} w_j \left\{d_{t,j} + \ln\left(e^{-d_{t,j}} + S_t\right)\right\}\right].$$

This has the log-softmax form of InfoNCE with negative squared-distance similarity. In standard predictive InfoNCE, both positive and negative similarities are evaluated relative to the predicted future state $\hat{z}_{t,j}$. In world2vec-DCL, negative similarities instead use the starting embedding $h(X_t)$ and therefore do not depend on the dynamics model. Consequently, the dynamics model receives only a positively rescaled prediction-error gradient, while the encoder additionally receives gradients encouraging separation from negative examples.

The authors also evaluate representation identifiability by testing whether independently trained encoders recover equivalent representations up to affine transformations:

$$\max_{A,b} R^2\left(h_2(X), Ah_1(X) + b\right).$$

This consistency measure is used for evaluation, not as a training constraint, and does not by itself establish recovery of the underlying physical state.

References

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